

ECON 311

Lecture 13: The IS-MP Model, part II

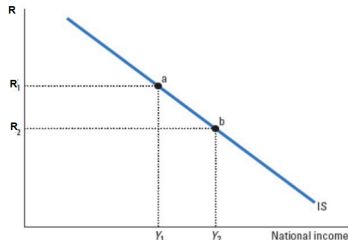
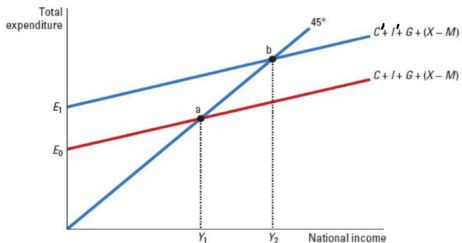
The Goods Market in the Short Run

Monetary Policy

Appendix

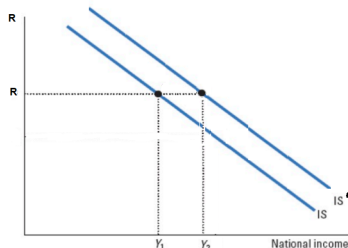
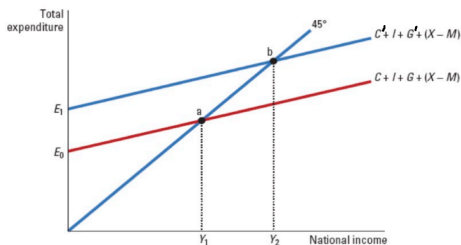
Deriving the IS-curve

$$\text{IS curve: } Y = \left[\frac{1}{1-b_c} \right] * \left[a_c + a_i - b_i(R) + \bar{G} + \bar{X} - \bar{M} \right]$$



Parametric Shifts in the IS-curve

$$\text{IS curve: } Y = \left[\frac{1}{1-b_c} \right] * \left[a_c + a_i - b_i(R) + \bar{G} + \bar{X} - \bar{M} \right]$$



Change of scale

- We are interested in $\tilde{Y}_t = \frac{Y_t - \bar{Y}}{\bar{Y}}$, so we are going to rewrite the equilibrium as follows:

$$\tilde{Y}_t = \left[\frac{1}{1 - b_c} \right] * [\bar{a} - \bar{b}_i(R_t - \bar{r})] - 1$$

$$\blacktriangleright \bar{a} = \frac{a_c}{\bar{Y}} + \frac{a_i}{\bar{Y}} - \frac{b_i}{\bar{Y}}\bar{r} + \frac{\bar{G}}{\bar{Y}} + \frac{\bar{X}}{\bar{Y}} - \frac{\bar{M}}{\bar{Y}}$$

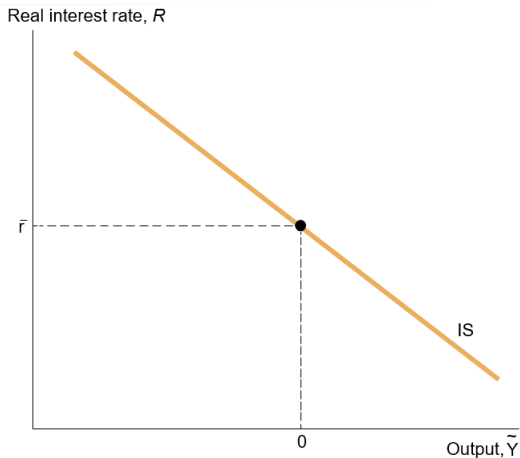
$$\blacktriangleright \bar{b}_i = \frac{b_i}{\bar{Y}}$$

- If all exogenous variables are at their long-run equilibrium:

$$\left[\frac{1}{1 - b_c} \right] * \bar{a} = 1 \Rightarrow \tilde{Y}_t = \left[\frac{1}{1 - b_c} \right] [-\bar{b}_i * (R_t - \bar{r})]$$

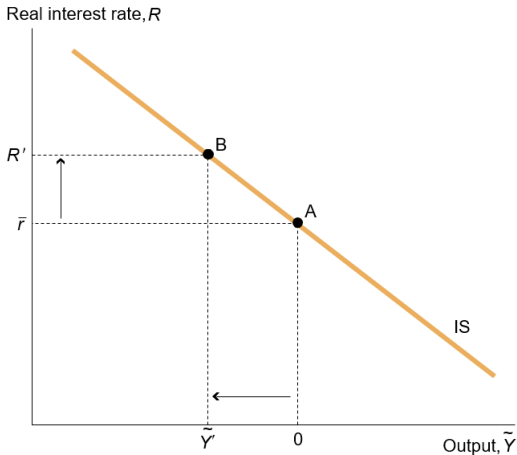
IS-curve with new scale

$$\text{IS-curve: } \tilde{Y}_t = \left[\frac{1}{1-b_c} \right] * \left[\bar{a} - \bar{b}_i(R_t - \bar{r}) \right] - 1, \quad \bar{a} \left[\frac{1}{1-b_c} \right] = 1$$



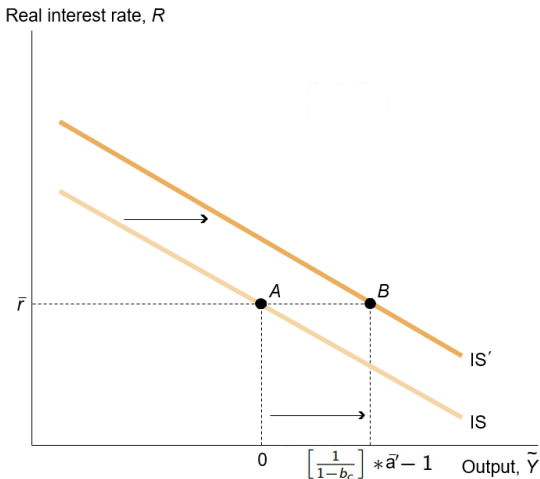
Changes in the real interest rate

$$\text{IS-curve: } \tilde{Y}_t = \left[\frac{1}{1-b_c} \right] * \left[\bar{a} - \bar{b}_i(R_t - \bar{r}) \right] - 1, \quad \bar{a} \left[\frac{1}{1-b_c} \right] = 1$$



Temporary Aggregate Demand Shock

$$\text{IS-curve: } \tilde{Y}_t = \left[\frac{1}{1-b_c} \right] * [\bar{a} - \bar{b}_i(R_t - \bar{r})] - 1, \quad \bar{a} \left[\frac{1}{1-b_c} \right] = 1, \quad \bar{a}' > (1 - b_c)$$

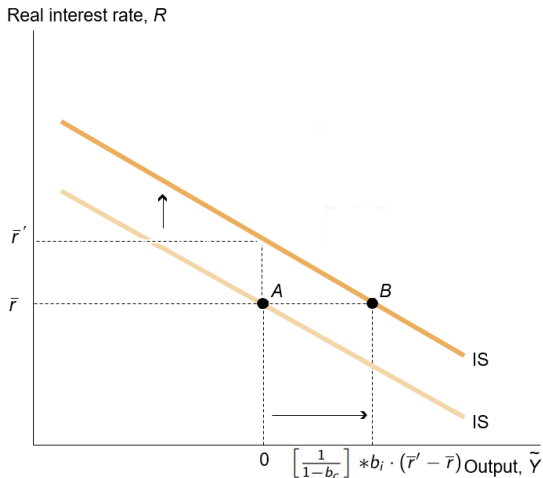


Permanent Aggregate Shock

- Permanent shocks impact potential output and originate from the production side of the economy, which have two effects:
 1. Permanent shocks affect demand parameters, which affect both long- and short-run output levels equally.
 2. Permanent production shocks affect the MPK
 - ▶ This has no effect in the long-run because $\bar{r} = MPK$
 - ▶ This results in output overshooting in the SR as the economy adjusts to its new potential output.

E.g. Permanent TFP shock

$$\text{IS-curve: } \tilde{Y}_t = \left[\frac{1}{1-b_c} \right] * [\bar{a} - \bar{b}_i(R_t - \bar{r})] - 1, \quad \bar{a} \left[\frac{1}{1-b_c} \right] = 1, \quad \bar{r}' > \bar{r}$$



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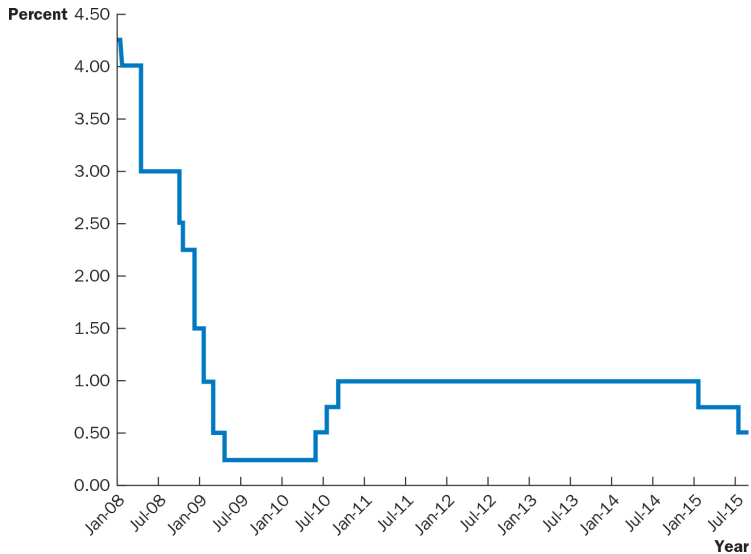
Monetary Policy

- The BoC currently has three main monetary policy variables at its disposal.
 1. Overnight rate
 2. Open-market operations
 3. Reserve requirements

Overnight rate

- Overnight rate: The interest rate on very short-term loans between commercial banks.
 - ▶ The BoC can indirectly change the interest rate at which commercial banks lend to each other.
- When banks lend too much and do not meet the reserve requirement, they are able to borrow short-term from other commercial banks which have excess reserves or from the BoC itself at the overnight rate to meet the requirement.
 - ▶ A high overnight rate encourage banks to lend less.
- The BoC announces changes in the overnight rate on eight fixed days each year (around every month and a half).

Overnight rate



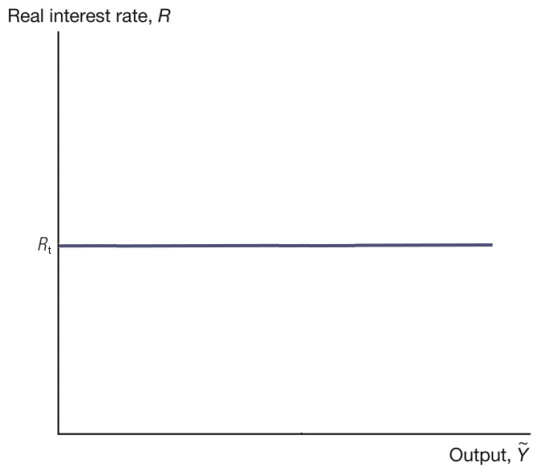
From nominal to real interest rate

- By modifying the overnight rate, the BoC affects the nominal interest rate in the market.
 - ▶ Commercial banks' additional borrowing costs is pass on to borrowers
- We can study the effect of changes in the nominal interest rate on the real interest rate using the fisher equation:

$$R_t = i_t - \pi_t$$

- ▶ We are going to assume that in the SR, prices are sticky and only change slowly over time.
- ▶ In the SR, $\Delta\pi$ do not completely offset Δi (this is different in the long-run).
- ▶ Therefore, in the SR, CB monetary policy has an indirect influence on the real rate return, R .

The MP curve

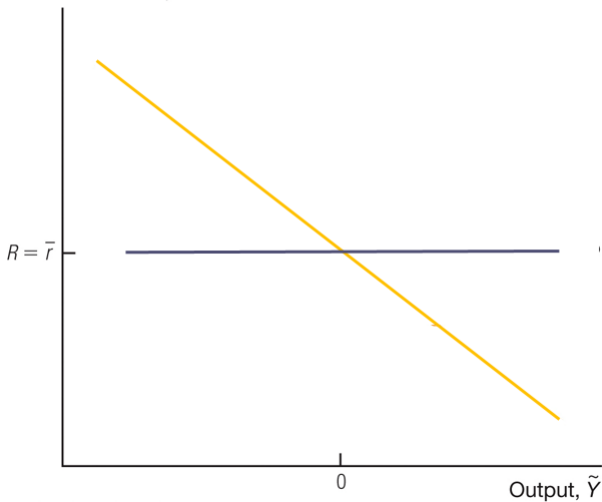


The IS-MP model

- The MP curve
 - ▶ Illustrates the central bank's ability to set the real interest rate at a particular value (horizontal line)
- The IS curve
 - ▶ Illustrates the negative relationship between short-run output and interest rates (through changes in aggregate expenditure)

The IS-MP model

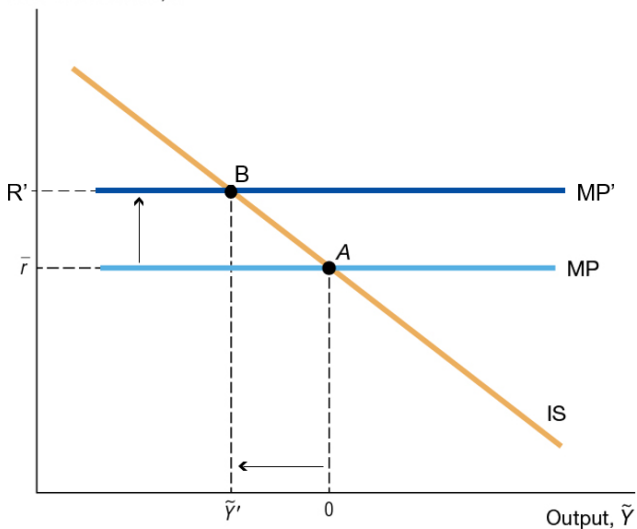
The MP Curve in the IS-MP Diagram
Real interest rate, R



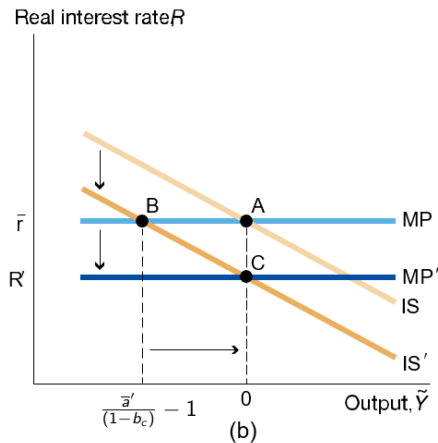
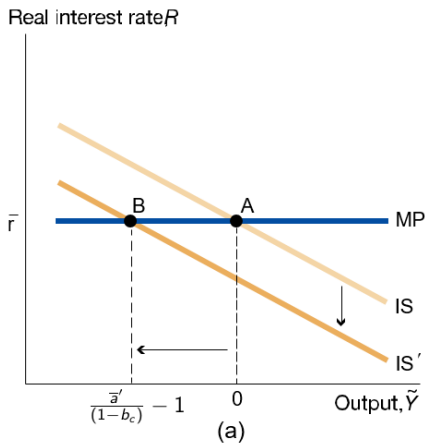
The IS-MP model

Raising the Interest Rate in the IS-MP Diagram

Real interest rate, R



Temporary Negative Demand Shock



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Appendix

Change of Scale: Derivation (part 1)

- We begin with the solution to Y_t in the SR equilibrium:

$$Y_t = \frac{1}{1 - b_c} (a_c + a_i - b_i R_t + \bar{G} + \bar{X} - \bar{M})$$

- We first divide both sides by the long-run output level $\bar{Y}$ and then subtract one from both sides:

$$\underbrace{\frac{Y_t}{\bar{Y}} - 1}_{\tilde{Y}_t} = \frac{1}{1 - b_c} \left(\frac{a_c}{\bar{Y}} + \frac{a_i}{\bar{Y}} - \frac{b_i}{\bar{Y}} R_t + \frac{\bar{G}}{\bar{Y}} + \frac{\bar{X}}{\bar{Y}} - \frac{\bar{M}}{\bar{Y}} \right) - 1$$

- We add and subtract $\frac{b_i}{\bar{Y}} \bar{r}$:

$$\tilde{Y}_t = \frac{1}{1 - b_c} \left(\frac{a_c}{\bar{Y}} + \frac{a_i}{\bar{Y}} - \frac{b_i}{\bar{Y}} R_t + \frac{b_i}{\bar{Y}} \bar{r} - \frac{b_i}{\bar{Y}} \bar{r} + \frac{\bar{G}}{\bar{Y}} + \frac{\bar{X}}{\bar{Y}} - \frac{\bar{M}}{\bar{Y}} \right) - 1$$

Change of Scale: Derivation (part 2)

- We rearrange terms to obtain:

$$\tilde{Y}_t = \frac{1}{1-b_c} [\bar{a} - \bar{b}_i (R_t - \bar{r})] - 1$$

where,

$$\bar{a} = \frac{a_c}{\bar{Y}} + \frac{a_i}{\bar{Y}} - \frac{b_i}{\bar{Y}} \bar{r} + \frac{\bar{G}}{\bar{Y}} + \frac{\bar{X}}{\bar{Y}} - \frac{\bar{M}}{\bar{Y}}$$

and,

$$\bar{b}_i = \frac{b_i}{\bar{Y}}$$

$$\text{and, } \bar{Y} = \frac{1}{1-b_c} \left(a_c^{LR} + a_i^{LR} - b_i^{LR} \bar{r} + \bar{G}^{LR} + \bar{X}^{LR} - \bar{M}^{LR} \right)$$

- From the previous equation, we can see that if all parameters in the SR are equal their long-run levels:

$$\bar{a} = \frac{a_c^{LR} + a_i^{LR} - b_i^{LR} \bar{r} + \bar{G}^{LR} + \bar{X}^{LR} - \bar{M}^{LR}}{\frac{1}{1-b_c} \left(a_c^{LR} + a_i^{LR} - b_i^{LR} \bar{r} + \bar{G}^{LR} + \bar{X}^{LR} - \bar{M}^{LR} \right)}$$

- Solving for this results in: $\bar{a} = 1 - b_c$